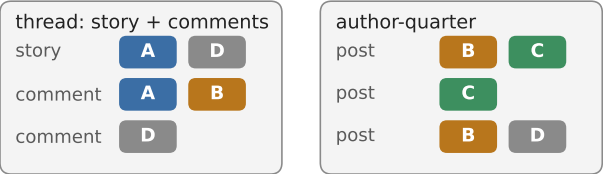
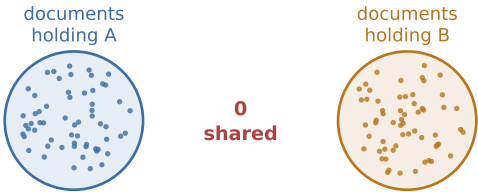


a The document, two ways: a thread, or an author-quarter **b** An eligible pair: should have met, never did

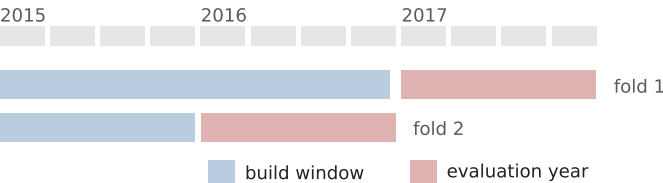


Two document definitions. A thread is a story with its top comments; an author-quarter is every concept one person used in three months. Two concepts co-occur when one document holds both.



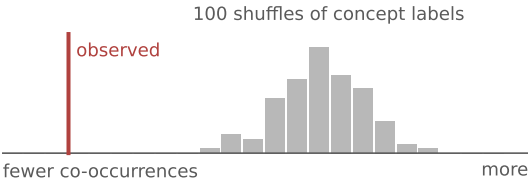
Both concepts are frequent. Had documents picked concepts independently, at least 2 build-window documents would hold both. Observed: none. The pair is eligible, or suppressed: it should have met and never did.

c Two folds: build on past years, evaluate the next



Eligible pairs are found in the build window; both readouts are measured in the evaluation year. Fold 1 builds on 2015 to 2016 and evaluates 2017; fold 2 builds on 2015 and evaluates 2016. Every fold boundary was registered before the evaluation ran.

d The segregation statistic: observed count vs. label shuffles



Pooled over all eligible pairs: the evaluation documents holding any of them, against 100 shuffles that keep every document's size and every concept's frequency. $z = (\text{observed} - \text{shuffle mean}) / \text{shuffle sd}$. z far below 0: the pairs are kept apart.

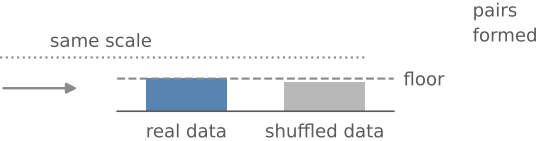
e Two criteria for 'this pair formed', and what shuffled data do under each

z-criterion (retired)



Expected count from each concept's own frequency. Large documents make joint counts combinatorially; shuffled data 'form' more pairs than the real data.

per-pair permutation criterion



Each pair's threshold is the 99th percentile of its own shuffled counts. Real and shuffled data both sit at the 1 percent false-positive floor.